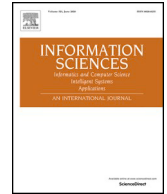




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## CFERE: Multi-type Chinese financial event relation extraction

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## ABSTRACT

Extracting various types of event relations in financial texts can benefit many downstream applications supporting financial analysis. This paper addresses the multi-type event relation extraction problem in the finance domain focusing on handling several issues in existing studies, including (1) limited event relation types involved, (2) insufficient feasibility when handling non-annotated data, (3) ineffectiveness in recognizing multi-type event relation, and (4) the asynchronous event and event relation extraction process. To tackle these limitations, we carefully define six types of event relations based on the characteristics of financial texts (e.g., abundant numerical words) and further devise an integral framework for Chinese financial event relation extraction. The framework is capable of handling unsupervised event extraction and event relation recognition jointly. Specifically, according to linguistic characteristics, a Core Verb Chain is employed for the event identification. Then, by constructing a Syntactic Semantic Dependency Parsing graph, scattered events are combined into pairs, and event ellipsis elements can be completed to prevent event information loss. Also, to capture more sentence semantics, we formulate an Event Restore module that converts the structured event pairs into event restore sentence pairs and pour the pairs into the BERT model for relation type identification. Finally, to enhance the embeddings for event core elements, an Event Core Embeddings layer is augmented in BERT, and we fine-tune the model on our annotated financial corpus. Extensive experiments are conducted to validate the effectiveness of our method, and the results confirm its superiority over the state-of-the-art baselines.

## 1. Introduction

Event relation extraction is an essential procedure in relation extraction, which significantly benefits natural language processing applications, such as information extraction [1–5], investment strategy [6,7], and question answering [8,9], etc. By capturing the correlations, associated events can complement each other and give full play to their values.

Financial articles and news usually report finance and economic-related activities, such as enterprises' operating conditions, stock movements, etc. For instance, a common structure of financial news can be as follows. It first introduces the general trend of the

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enterprises or the stock market, such as the rise or fall, followed by specific numerical values. Without further explanations, it is difficult for readers to clearly understand the essential idea behind the numbers. For example, whether a value indicates a good or bad situation should depend on an over-time or cross comparison. Such discussions may result in abundant relations (e.g., *Supplement* and *Degree*). Fig. 1 shows the supplement relation between events  $ET_{1-1}$  and  $ET_{1-2}$ , and the degree modification relation between events  $ET_{2-1}$  and  $ET_{2-2}$ . Note that, an event is represented by  $ET(Sub, Pred, Obj)$ , and  $ET(\cdot)$  denotes the core element set of a given event; *Sub*, *Pred*, and *Obj* denote subject, predicate, and object, respectively.

Furthermore, more extensive descriptions regarding a company may also be found in financial articles, such as (1) the current business situation along with the plan, (2) a certain action leading to a certain result, (3) different aspects of the company, and (4) a series of behaviors of enterprises under certain adversity or adverse factors, and the like. These cases lead to a large number of relation types, including *Strategy* (a strategy/plan based on the current situation, such as events  $ET_{3-1}$  and  $ET_{3-2}$ ), *Causality* ( $ET_{5-1}$  and  $ET_{5-2}$ ), *Parataxis* ( $ET_{6-1}$  and  $ET_{6-2}$ ) and *Contrast* ( $ET_{7-1}$  and  $ET_{7-2}$ ). When articles use specific companies as instances to describe the overall market state, we may see numerous *Phenomenon-Instance* relations ( $ET_{4-1}$  and  $ET_{4-2}$ ). A detailed discussion of relation types can be found in Section 3.

Compared with other fields, financial texts usually contain more event relation types. Given the complexity of financial events and their essential social impacts, our work seeks solutions to the extraction of multi-type finance-related event relations. Note that, “multi-type” refers to event relation types with different category meanings, such as causality, supplement relation, contrast relation, and so on.

Although event relation extraction and related topics have been extensively studied, threefold issues are still under exploration. First, existing models only consider a small number of relation types, such as causal and temporal relation [10–12]. Although Hong et al. [13] presented five correlation types, they mainly focused on identifying the connection and hierarchical relations between events and sub-events.

Second, previous works primarily addressed fine-grained problems of a single relation type, such as implicit causality [14], document-level causality [10], and domain causality [15], which are tailor-made for the specific issue and unsuitable for the identification of multi-type event relations. For example, Zhao et al. [10] identified document-level causality via graph inference mechanism, while this mechanism is not suitable for non-causal event relation types. Also, existing studies are basically supervised models that require a large number of annotated training data (including event information), so they may not perform well in the finance domain given the limited annotated corpus. Last, most state-of-the-art techniques are developed based on English corpus, which may result in a weak performance when implemented in a scenario with other languages. Especially, Chinese is a paratactic language with vast short sentences and verbs (e.g., multiple events in one sentence, element ellipsis) [16], leading to a more challenging event extraction task.

Inspired by the powerful semantic representation of Bidirectional Encoder Representations from Transformers (BERT) [17], as well as its function for next-sentence prediction (i.e., relation recognition between contextual sentences), we consider the BERT model to identify our multi-type event relations. However, due to more than one event pair in a sentence, it does not work for BERT to identify event relation type if directly feeding sentences. Therefore, events must be extracted first, and then judge the relation types of event pairs via BERT.

To address the aforementioned problems, we propose a Chinese Financial Event Relation Extraction framework (CFERE). First, based on corpus features and the characteristics and finance domain, we define six event relation types, including *Supplement*, *Degree*, *Strategy*, *Phenomenon-Instance*, *Parataxis*, *Contrast*, consisting of seven relation types with event causality. Then, considering the scarcity of annotated data, we sort to an unsupervised method [16] to implement event extraction, which not only prevents event annotation, but also establishes the semantic association for extracted events, forming event pairs for subsequent relation recognition. Moreover, this method performs the completion for omitted event elements to improve the quality of event extraction.

Nevertheless, we discover the events extracted by literature [16] miss some significant information (e.g., adversative and negative words), which can enhance the relation recognition. Thus, an Event Restore module is developed to restore structured events into event restore sentences and pour the associated event sentence pairs into BERT for fine-tuning or relation type recognition directly. Furthermore, event core elements (subject, predicate, and object) are essential to understand the semantic meanings of events for humans and are not explicitly tagged. Thus, we optimize the embedding encoding process of BERT. Based on BERT's original three-layer embeddings, we add an additional Event Core Embedding layer to handle event core elements, constituting a four-layer embedding to jointly encode the inputs of BERT.

To sum up, the contributions of our work can be summarized as follows.

- To achieve a comprehensive event relation extraction in finance, we define six event relation types based on practical needs in the financial domain.
- Following the definitions, the CFERE framework is explored to extract multi-type event relations.
- To provide more significant information and encode event core elements, BERT's inputs and internal embeddings are optimized.
- Extensive experiments are conducted to evaluate the effectiveness of our framework, and the results demonstrate a consistent superiority of our method over state-of-the-art techniques.

The remainder of this paper is structured as it follows. First, we review related work on event and event relation extraction in Section 2. Next, Section 3 introduces the formal definition of six event relation types, followed by detailed illustration of our method in Section 4. After that, Section 5 demonstrates the experimental settings and key results. In-depth discussions on our results are further provided in Section 6. Finally, Section 7 concludes the paper.

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sentences | <p>[S<sub>1</sub>]: 青岛银行 2018 年净利为 2033 万元, 同比增长 6.48%.<br/>(Qingdao bank's net profit in 2018 was 20.33 million Yuan, grew 6.48% year-on-year.)</p> <p>[S<sub>2</sub>]: 中原银行下跌, 跌破 2300 关口, 累跌 20%, 创上市新低, 暂连跌三日.<br/>(ZhongYuan Bank dropped, fell below the 2300 mark, fell in total 20%, hit a listed new low, temporarily fell for three consecutive days.)</p> <p>[S<sub>3</sub>]: 康臣药业飙升 9%, 拟回购 4500 万股.<br/>(ConSun surged nearly 9%, planned to buy back 45 million shares.)</p> <p>[S<sub>4</sub>]: 午后猪肉概念股拉升, 新希望大涨超 8%.<br/>(Afternoon Pork Concept Stock pulled up, New Hope rose over 8%.)</p> <p>[S<sub>5</sub>]: 金蝶国际升逾 2%, 获瑞信上调目标价 8%.<br/>(Kingdee rose over 2%, credited Resun raising the target price 8%.)</p> <p>[S<sub>6</sub>]: 碧桂园跌超 4%, 昨日回购 1800 万股.<br/>(BiGuiYuan fell over 4%, bought back 18 million shares yesterday.)</p> <p>[S<sub>7</sub>]: 台海核电终于要分红了, 但股市却不买账.<br/>(TaiHai Nuclear is finally paying dividends, but the stock market does not buy it.)</p>                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Events    | <p>[S<sub>1</sub>]: <math>ET_{1-1}</math>(青岛银行净利, 为, 2033 万元), <math>ET_{1-2}</math>(, 增长, 6.48%)<br/>(Qingdao bank's net profit, was, 20.33 million Yuan), (, grew, 6.48%)</p> <p>[S<sub>2</sub>]: <math>ET_{2-1}</math>(中原银行, 下跌, ), <math>ET_{2-2}</math>(, 跌破, 2300 关口), <math>ET_{2-3}</math>(, 累跌, 20%)<br/>(ZhongYuan Bank, dropped, ), (, fell below, the 2300 mark), (, fell in total, 20%)<br/><math>ET_{2-4}</math>(, 创, 上市新低), <math>ET_{2-5}</math>(, 连跌, 三日)<br/>(, hit, a listed new low), (, fell for, three consecutive days)</p> <p>[S<sub>3</sub>]: <math>ET_{3-1}</math>(康臣药业, 飙升, 近 9%), <math>ET_{3-2}</math>(, 拟, 回购 4500 万股)<br/>(ConSun, surged, nearly 9%), (, planned to, buy back 45 million shares)</p> <p>[S<sub>4</sub>]: <math>ET_{4-1}</math>(猪肉概念股, 拉升, ), <math>ET_{4-2}</math>(新希望, 大涨, 超 8%)<br/>(Pork Concept Stock, pulled up, ), (New Hope, rose, over 8%)</p> <p>[S<sub>5</sub>]: <math>ET_{5-1}</math>(金蝶国际, 升逾, 2%), <math>ET_{5-2}</math>(, 获, 上调目标价 8%)<br/>(Kingdee, rose over, 2%), (, credited, raising the target price 8%)</p> <p>[S<sub>6</sub>]: <math>ET_{6-1}</math>(碧桂园, 跌超, 4%), <math>ET_{6-2}</math>(, 回购, 1800 万股)<br/>(BiGuiYuan, fell over, 4%), (, bought back, 18 million shares)</p> <p>[S<sub>7</sub>]: <math>ET_{7-1}</math>(台海核电, 分红, ), <math>ET_{7-2}</math>(股市, 买账, )<br/>(TaiHai Nuclear, paying, dividends), (the stock market, buy, it)</p> |
| Relations | <p>[S<sub>1</sub>]: <math>ET_{1-1}</math>(青岛银行净利, 为, 2033 万元), <math>ET_{1-2}</math>(, 增长, 6.48%) Supplement</p> <p>[S<sub>2</sub>]: <math>ET_{2-1}</math>(中原银行, 下跌, ), <math>ET_{2-2}</math>(, 跌破, 2300 关口) Degree<br/><math>ET_{2-1}</math>(中原银行, 下跌, ), <math>ET_{2-3}</math>(, 累跌, 20%) Degree<br/><math>ET_{2-2}</math>(, 跌破, 2300 关口), <math>ET_{2-4}</math>(, 创, 新低) Supplement<br/><math>ET_{2-4}</math>(, 创, 新低), <math>ET_{2-5}</math>(, 连跌, 三日) Parataxis</p> <p>[S<sub>3</sub>]: <math>ET_{3-1}</math>(康臣药业, 飙升, 近 9%), <math>ET_{3-2}</math>(, 拟, 回购 4500 万股) Strategy</p> <p>[S<sub>4</sub>]: <math>ET_{4-1}</math>(猪肉概念股, 拉升, ), <math>ET_{4-2}</math>(新希望, 大涨, 超 8%) Phenomenon-Instance</p> <p>[S<sub>5</sub>]: <math>ET_{5-1}</math>(金蝶国际, 升逾, 2%), <math>ET_{5-2}</math>(, 获, 上调目标价 8%) Causality<sup>1</sup></p> <p>[S<sub>6</sub>]: <math>ET_{6-1}</math>(碧桂园, 跌超, 4%), <math>ET_{6-2}</math>(, 回购, 1800 万股) Parataxis</p> <p>[S<sub>7</sub>]: <math>ET_{7-1}</math>(台海核电, 分红, ), <math>ET_{7-2}</math>(股市, 买账, ) Contrast</p>                                                                                                                                                                                                                                                                                                                                                                  |

<sup>1</sup> To conveniently illustrate and understand, we also give an example for existing event causality.

Fig. 1. Examples of financial news headlines, including events and corresponding event relations.

## 2. Related work

Related studies of this work fall into two categories, including event extraction and event relation extraction.

### 2.1. Event extraction

As the basis of event relation extraction, event extraction models have been extensively studied [18,19]. Herein, we only review the efforts that address the lack of data, which is related to our method, mainly involving the following four aspects. (1) **Automatically labeled data generation.** It is commonly realized via the distant supervision method and mapping external knowledge base (e.g., Freebase, FrameNet) to the targeted event structure [20]. (2) **Transfer learning.** Transform original task to other problems with abundant data, or train models in other task/domain and then fine-tune on target task/domain. Liu et al. [21] studied a new learning paradigm of event extraction by explicitly casting it as a Machine Read Comprehension problem, in which there are large-scale datasets. (3) **Unsupervised method.** Numerous researchers explored clustering methods to extract events, as well as statistical models. For example, Shen et al. [22] represented each event type as a cluster of <predicate sense, object head> pairs. Some others investigated the extraction task with pattern-matching schemes based on the characteristics of language and domain [16]. (4) **Few-shot/zero-shot learning.** These technologies explore new models that are suitable for fewer samples [23].

The limitations of the above methods when extracting events in Chinese financial texts are summarized as follows. First, external knowledge bases are required. Second, without association between events, subsequent event relation recognition is a difficult task. Third, existing methods mainly focus on English and general corpus without considering specific domains. Last, existing models only focus on some specific event types, which are uninformative for event relation recognition. Considering the characteristics of finance (e.g., rich in numerical words and company names) and the Chinese language (e.g., many ellipses), we employ the unsupervised method proposed in [16].

### 2.2. Event relation extraction

Regarding event relation extraction, most of the works focused on causality and temporal relation. In the following, we review the studies of the two relation types.

To extract the causality relation, machine learning methods have been extensively used. Kruengkrai et al. [14] employed a multi-column convolutional neural network to handle multiple background knowledge sources for the causality that lacks explicit clues indicating the existence of event causality. Nordon et al. [15] built a causality graph based on electronic medical records for the effective analysis of dependencies in these data. Different ideas for handling document-level event causality have also been developed, such as encoded lexical and syntactic features [24], event networks [11], and graph inference mechanism-based extraction [10].

Regarding event temporal relation, previous studies were mainly based on a template or feature engineering (e.g., lexical and syntactic features). Ng et al. [25] developed a hybrid system combined with a classifier containing 437 hand-coded rules. Mirza and Tonelli [26] described an approach to label temporal links using simple features based on lexico-syntactic information. Subsequently, Choubey and Huang [27] designed a sequence model to extract 14 temporal relations on the Timebank corpus. Ning et al. [28] presented a joint inference framework for causal and temporal relations using constrained conditional models. Subsequently, Vashishtha et al. [29] argued a semantic framework to handle fine-grained event temporal relations. Junuthula et al. [30] proposed a block point process model for dynamic networks evolving in the form of timestamped events.

In summary, existing studies are limited in three ways. First, fine-grained problems of single-event relations are considered, such as implicit relations and document-level causality. Since these methods concern specific relations and problems, they can be ineffective when dealing with multiple event relation types. For instance, the graph inference mechanism proposed by [10] is unsuitable for non-causal event relation types, such as the supplement relation that focuses on information supplementation rather than the semantic meaning of causal inference. Second, supervised models may be inefficient in the finance field due to insufficient annotated data for training. Last, event relation recognition is separated from event extraction. When implementing event relation extraction, the information of events is usually assumed to be given/tagged, resulting in the task being divided into two independent parts without a unified extraction framework.

## 3. Chinese financial event relations

To facilitate the understanding of event relations, we still use the examples in Fig. 1 to describe our six new types of event relations, consisting of seven event relations with existing event causality.

### 3.1. Seven event relations

*Supplement* is a relation that an event depicts another event element in more detail. This is a prominent event relation type in finance, often triggered by numerical words. Establishing *Supplement* can make event-related statements more convincing and precise. As shown in line 1 of Fig. 1, the relation type between events  $ET_{i-1}$  and  $ET_{i-2}$  is *Supplement*, given that the content of  $ET_{i-2}$  is mainly to supplement the “20.33 million yuan” in  $ET_{i-1}$ . Thus, readers can perceive the strength of “20.33 million yuan” (i.e., the value is more or less?).

The *Degree* relation refers to cases in which an event describes the strength level of another event that is generally associated with the predicate of an event. *Degree* describes the degree of an event behavior, so that the effect of the event can be better measured, leading to a more persuasive event description. For instance, event  $ET_{2-1}$  in Fig. 1 reveals the “dropped” behavior, while the extent is described in events  $ET_{2-2}$  and  $ET_{2-3}$ , respectively, such as “the 2300 mark” and “20%”. In this paper, the relation of one event modifying another event behavior is named *Degree*.

The *Strategy* relation states the situation that a strategy event will be formulated based on another event, in which the strategy event may happen but did not. Establishing this relation can help predict a possible event that has not occurred but may happen. The user can realize the probability of strategy events for adjusting the weight of events used for economic prediction. For example, given a company’s annual report, the significance of a forecast is not the same as a formal report for investors. In Fig. 1,  $ET_{3-2}$  is the strategy event formulated by the company according to the existence of event  $ET_{3-1}$  and business requirements. Note that, event  $ET_{3-2}$  did not occur and was only a plan. Thus, the relation type between them is defined as *Strategy*. Along this line,  $ET_{3-2}$  should be less significant than  $ET_{3-1}$ .

The *Phenomenon-Instance* relation is the situation that one event describes the overall environment of industry and another depicts a specific instance. By instance events, the phenomenon events describing the general trend of the market are made more specific and convincing. Instead, instance events can learn their industry context with phenomenon events. For example, event  $ET_{4-1}$  describes the overall situation of “Pork Concept Stock”, while event  $ET_{4-2}$  states a more specific condition of the stock “New Hope”, which rises more than 8% on the background of  $ET_{4-1}$ . The identification of this relation can result in a more precise description of events. Note that, the behaviors of instance events should be consistent with phenomenon events (i.e., both increase or decrease). If the behaviors are opposite, the type is classified as *Contrast*.

The *Causality* (causal relation) is identified when an event causes another. Result events may have different causes; thus, the practical impacts of a result event may vary. *Causality* can be applied to many scenarios (e.g., further studies regarding inference). In finance, it is common that one type of behavior triggers another, such as events  $ET_{5-1}$  and  $ET_{5-2}$ .  $ET_{5-1}$  reflects the “rose over” behavior of “Kingdee”, and the extent attains “2%”. Because of  $ET_{5-1}$ , event  $ET_{5-2}$  happens (i.e., “credited Resun raising the target price 8%”). Since they have a strong semantic cause and result, we regard such relations as *Causality*.

The *Parataxis* relation refers to a situation that different events describe multiple aspects of a subject, which is beneficial to drawing comprehensive pictures for enterprises. Given a specific enterprise, whether leaders of the company or external investors, they all need to have a comprehensive view of the enterprise. The sentence  $S_6$  in Fig. 1 demonstrates two independent aspects (“fell over” and “bought back”) of “BiGuiYuan”, a real estate company in China, using events  $ET_{6-1}$  and  $ET_{6-2}$ . Therefore, the type is *Parataxis*.

The *Contrast* relation is mainly used to depict the situation that semantic meanings between events that are opposite or contradictory. The establishment of such a relationship can obtain the adverse factors or disadvantages of associated events, assisting event users in analyzing the probability and regulating the weight of events in the prediction application. In Fig. 1, event  $ET_{7-1}$  reveals the “paying” behavior, while the semantics of event  $ET_{7-2}$  is opposite demonstrated by “but the stock market does not but it”. Therefore, the relation type should be *Contrast*. Without this correlation, the significance of event  $ET_{7-2}$  used alone drops sharply or is even useless due to the lack of description objects. Also, event  $ET_{7-1}$  cannot be aware of the adverse factors of “TaiHai Nuclear” for dividend, inflating the application significance to a certain extent. However, the connective “but” is missing in event  $ET_{7-2}$  and it is crucial to the recognition of *Contrast* relation. Similarly, the negative word “not” is also important. Thus, such words should be included in the inputs of the relational recognizer to strengthen its identification ability.

### 3.2. Dealing with ambiguous event relations

In the following, we clarify the differences between four pairs that are easily confused in the seven types of event relations.

- *Supplement & Degree*: *Supplement* aims to give a statement for elements of other events in detail, while *Degree* focuses on the extent modifies of event behaviors. For example, event  $ET_{1-2}$  complements the element “20.33 million Yuan” of event  $ET_{1-1}$ , and event  $ET_{2-2}$  reveals the degree of “dropped” behavior in event  $ET_{2-1}$ .
- *Degree & Parataxis*: There is no degree modification between events in *Parataxis*, and it tends to describe different aspects of the same subject. For instance, event  $ET_{6-1}$  in Fig. 1 presents the “fell over” behavior, and the strength “4%” is also in the event. Moreover, event  $ET_{6-2}$  is not a further description of this behavior. Therefore, the relation type is *Parataxis*.
- *Phenomenon-Instance & Causality*: *Phenomenon-Instance* emphasizes the subordinate relation in which the instance event is a specific example of the phenomenon event. Regarding *Causality*, there is a strong logic of cause and result in semantics, and the subjects of events basically do not have a subordinate relation. Specifically, the “rose” behavior in event  $ET_{4-2}$  of Fig. 1 is not caused by event  $ET_{4-1}$ , but indicates that it is formed in the environment of event  $ET_{4-1}$ . Thus, we classify it as *Phenomenon-Instance*. In contrast, event  $ET_{5-2}$  is semantically caused by the “rose over” behavior in event  $ET_{5-1}$ , and it is not an instance of  $ET_{5-1}$ , thus the relation type is considered as *Causality*.
- *Strategy & Causality*: *Strategy* is defined to represent the relation of formulating a strategy event based on another event; that is, a strategy event is highly related to other events, and there is no significant causal semantic meaning between them. Also, *Strategy* focuses on subsequent planning, and there are events that did not happen, either one or two. In *Causality*, events must have happened.

## 4. Methodology

In this section, we describe our CFERE framework, as shown in Fig. 2. It has five major components (green background): (1) the Core Verb Chain, for identifying events from the syntactic Dependency Parsing (DP) tree, where red nodes are event predicates and blue nodes denote their subjects or objects; (2) the Syntactic Semantic Dependency Parsing (SSDP) graph, handling event extraction, the association between events (pink dotted line), and the completion of omitted event elements (blue dotted line); (3) the Event Restore, producing Event Restore Sentences (ERS) for extracted events; (4) the Event Core Embeddings (ECE) layer, functional in encoding event core elements (purple background boxes at the bottom); (5) the Relation Classifier, for predicting the event relation type directly or by fine-tuning on the financial corpus.

In addition, Fig. 2 also demonstrates DP and Semantic Dependency Parsing (SDP) trees generated by the syntactic parsing tool LTP.<sup>1</sup> On the bottom, it also shows the BERT's original three embedding layers: token, segment, and position.

The CFERE implements as follows. First, given a sentence, the DP and SDP trees are obtained. Then, we add the semantic dependency in SDP tree to the DP tree and adjust the structure of DP tree according to the Core Verb Chain mechanism, forming the final SSDP graph. Based on the SSDP graph, event extraction and ellipsis completion can be performed (see examples 1 and 2 in Fig. 2). Events  $ET_{7-1}$  and  $ET_{7-2}$  are extracted from sentence  $S_7$  in example 1, and the omitted element “Zhongyuan Bank” (blue dotted line) is completed in the last four events of example 2. Subsequently, we restore the extracted events to event restore sentences and combine them into event restore sentence pairs according to the edges between events' predicates. Finally, each pair is fed to the optimized BERT for relation type prediction by the Relation Classifier. The whole framework can work without annotated corpus in the event extraction process. Given a specific domain, we only annotate event relation types for fine-tuning to achieve better performance.

### 4.1. Core verb chain

Based on the syntactic parsing tool, a sentence could be converted to a dependency parsing tree, called DP tree, where nodes represent words, and edges are syntactic dependency relations. Fig. 3 demonstrates the DP tree of  $S_2$  in Fig. 1. For the sake of understanding, words before and after their parent are the left and right children in all figures of DP trees, according to their orders in original sentences. Moreover, all nodes are numbered in ascending order based on the above order, and the Root node id is 0.

Although the DP tree outputs one core word, which is not conducive to extracting events in a sentence, multiple core verbs could be identified based on dependency relation and part-of-speech. For example, in Fig. 3, sentence  $S_2$  contains five events triggered by verbs “dropped”, “fell below”, “fell in total”, “hit”, and “fell for”. The first has been marked as a core word by DP parsing, while the others are in two chains, in which the dependency relation and part-of-speech of nodes are COO and verb. The specific chains are “dropped”→“fell below”→“hit”→“fell for” and “dropped”→“fell in total”.

In linguistics, parallel predicates should be associated with some parallel symbols in the syntactic structure (e.g., the dependency relation COO used here). With these clues (verb chains and parallel structure form), all predicates of events can be detected. Importantly, the dependency relation must be continuous from the core word node to the current node; that is, there can be no other dependency relations in the path between them. Such chains are defined as Core Verb Chain (CVC), and nodes in it are called core verbs in reference [16].

Theoretically, the identification of event predicates should be 100% based on CVC, although the results may vary in different DP tools. Each node in CVC is a predicate that triggers an event, and its corresponding arguments depend on the predicate. Therefore, the event maps to a dependency subtree with the predicate as the root node, and the extraction of the event subject and object can be further performed by scanning the subtree.

### 4.2. Syntactic semantic dependency parsing

To extract events and complete ellipsis element, Wan et al. [16] added the semantic dependency to the DP tree and adjusted the dependency structure, constructing a Syntactic Semantic Dependency Parsing (SSDP) tree. Given that errors may exist in SDP parsing, only the semantic dependency between predicates is processed to reduce errors and inconsistency of syntactic and semantic dependency. Fig. 4 shows the SSDP tree of the sentence in Fig. 3.

However, although the SSDP tree contains syntactic and semantic information, extracting events and completing ellipsis elements are still challenging, because SSDP is not event-oriented, and the levels of nodes with the same status in events are different. For example, event “fell below” in Fig. 4 depends on event “dropped”, resulting in different levels for nodes in the two events. The level of “ZhongYuan Bank” node is higher than “mark”, “new low”, “fell for”, etc. This prevents the completion of event ellipsis element (e.g., “ZhongYuan Bank”), because event elements that belong to the same attributes should be at the same/adjacent level (e.g., subjects of an event should remain at the same/adjacent levels). Note that, “adjacent” refers to one or two levels below the current ellipsis element node.

To relieve these problems, all core verbs are tuned to the children of the Root node, ensuring they are on the same level and the same structural elements of different events as possible. Since the adjusted SSDP tree does not meet the definition of a tree, we name it the SSDP graph. Fig. 5 demonstrates the SSDP graph adjusted based on Fig. 4. The dependency relations between events represented by pink dotted lines are switched from direct to indirect.

<sup>1</sup> [http://www.ltp-cloud.com/intro\\_en](http://www.ltp-cloud.com/intro_en).

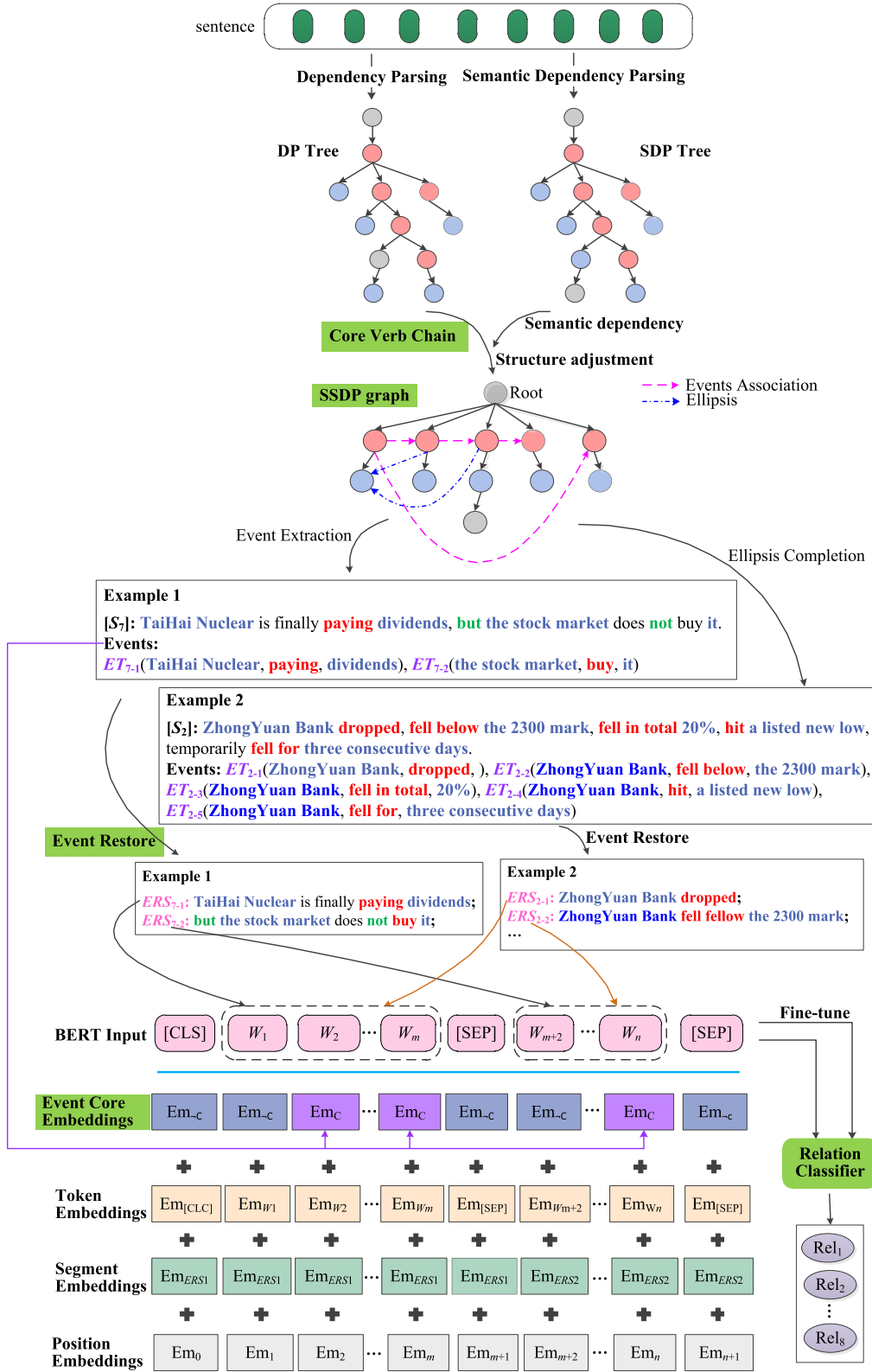


Fig. 2. The overall architecture of the CFERE framework.

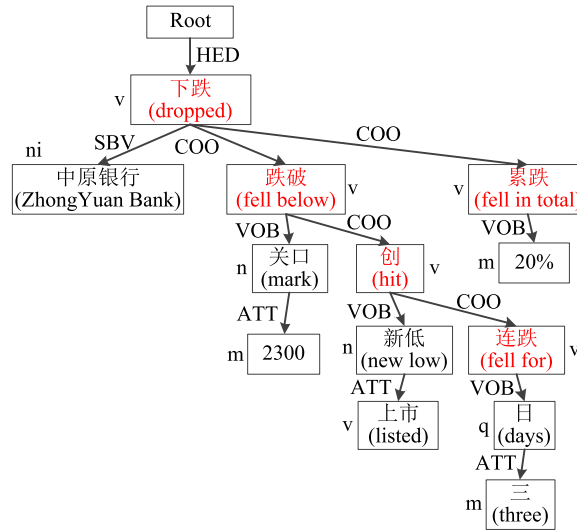


Fig. 3. The DP tree of  $S_2$  is parsed by the LTP tool. HED, COO, SBV, VOB, and ATT represent the dependency relations of the head, coordination, subject-verb, verb-object, and attributive, respectively. ni, n, v, m, and q are part-of-speech of organization name, general noun, verb, number, and quantity.

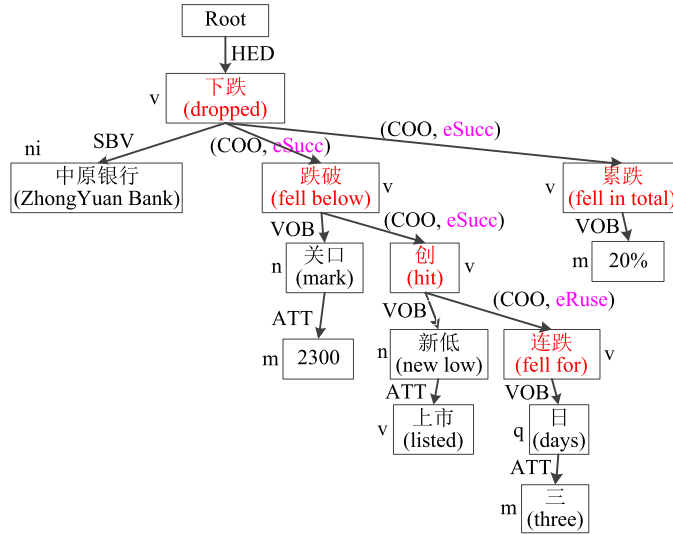


Fig. 4. SSDP tree of the sentence in Fig. 3. eSucc and eRuse denote semantic dependency of succession and result, respectively. Only dependency relations between predicates are given.

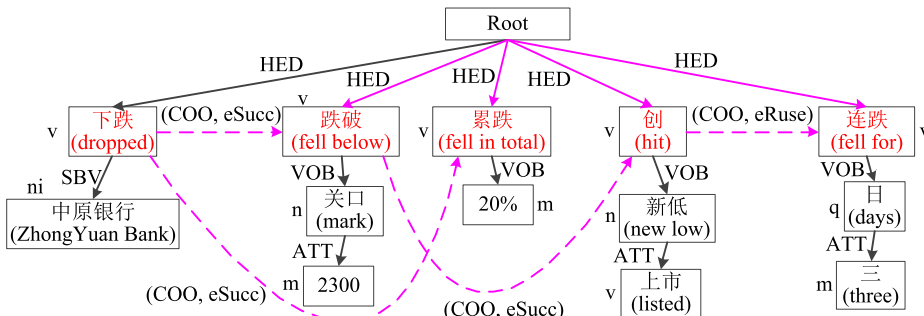


Fig. 5. The adjusted SSDP graph based on Fig. 4.

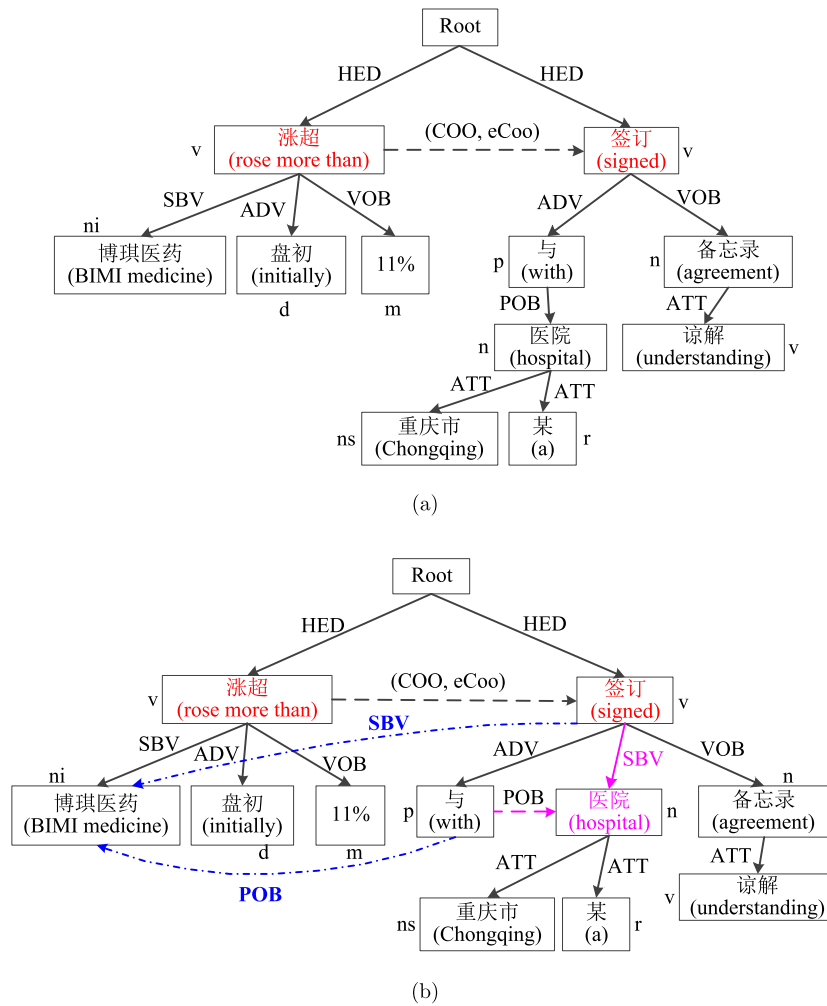


Fig. 6. Example of non-core verb adjustment. (a) is original SSDP tree of sentence  $S_8$ : “BIMI medicine initially rose more than 11% and signed understanding agreement with a Chongqing hospital”. (b) is the adjusted SSDP graph based on  $S_8$  in (a). POB and ADV are preposition-object and adverbial dependency relations, and eCOO denotes coordination.

Furthermore, the adjustment process of the SSDP tree also involves partial regulation for non-core verb nodes. For instance, objects depending on prepositions are moving to the corresponding core verb. Specifically, when nodes semantically depend on core verbs and the semantic relations are subject class (including Agt, Exp, Aft, Poss) or object class (including Pat, Cont, Prod, Orig, Datv),<sup>2</sup> they are adjusted to children of their core verb. The dependency relations of new edges for subject and object classes are SBV and VOB, respectively. Meanwhile, original dependency relations remain. Fig. 6(a) is the original SSDP tree of sentence  $S_8$ , which is transformed to Fig. 6(b) due to the subject semantic word “hospital” (pink line and font).

In addition, the element ellipsis is common in Chinese, especially the subject [16]. At present, there is no unified standard for the types of ellipses. Herein, we adopt the four subject ellipsis types in literature [16].

- *Simple Ellipsis*: the ellipsis event does not have a subject that is served by a simple element, e.g., event  $ET_{2-2}$  triggered by “fell below” in Fig. 5 omits the subject “ZhongYuan Bank”, which is a simple element in  $ET_{2-1}$  triggered by “dropped”, thus the type belongs to *Simple Ellipsis*.
- *Combined Ellipsis*: the ellipsis event does not have a subject, and a combination of elements is the ellipsis subject. For example, a whole event serves as the ellipsis element of another event.
- *Preposition Ellipsis*: partial subject ellipsis caused by prepositions, such as partial subject “BIMI medicine” being omitted due to preposition “with” in Fig. 6(b).
- *Indirect Ellipsis*: the ellipsis event has subjects but loses the modifiers. For example, sentence  $S_9$ : “KunLun Energy rose more than 3%, medium term net profit increased slightly by nearly 2%” contains events  $ET_{0-1}$  (KunLun Energy, rose, more than 3%) and

<sup>2</sup> [http://www.ltp-cloud.com/intro\\_en#sdp](http://www.ltp-cloud.com/intro_en#sdp) how.

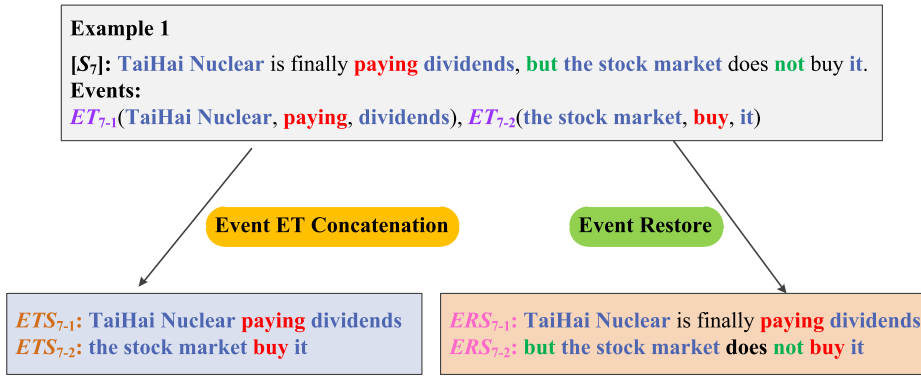


Fig. 7. Differences of event ET concatenation and event restoration for example 1 in Fig. 2.

$ET_{9-2}$  (medium term net profit, increased by, nearly 2%), where the subject “medium term net profit” of  $ET_{9-2}$  omits the modifier “Kun-Lun Energy” in  $ET_{9-1}$ .

According to Chinese language characteristics, we know that omitted elements are usually in previous events, while the SSDP graph not only maintains the order of words in the original sentence, but also ensures uniform structural words at the same level. Therefore, the completion of the ellipsis can be effectively implemented based on it. The main idea is to seek the related elements at the same level or deeper than the omitted subject in previous events associated with the omitted event. The completion strategies adopted in this paper are derived from [16]. In order to present a specific process of completion, we only introduce the completion method of common *Simple Ellipsis*. The steps are as follows: (1) search the subject in associated closest previous events, whose level is as same as the omitted element; (2) build a dependency edge between it and the event core verb existing ellipsis, such as “BIMI medicine” and “signed” in Fig. 6(b). Also, since the preposition “with” is parallel conjunction, a POB dependency relation needs to be built between “BIMI medicine” and “with”.

#### 4.3. BERT optimization

BERT is a pre-trained language model proposed by the Google team. At present, it has been used in a variety of tasks and achieves excellent results. Given that the Next Sentence Prediction task of BERT is similar to our goal, which aims to understand contextual relations for input sentence pairs, we select BERT as the classifier. However, our goal is to recognize multi-type event relations, rather than simple next-sentence analysis. Also, multiple events and event relations are contained in a sentence [31], which results in a failure to identify the relation types in a sentence. Given the sentence  $S_2$  in example 2 of Fig. 2, there are five events extracted based on the SSDP graph, and they can consist of four event pairs. Nevertheless, only one relation type is predicted by BERT if the input is the whole sentence, which does not correspond to multiple relation pairs. It means that BERT cannot directly work for us when feeding raw sentences/sentence pairs, so as to must input event pairs instead of them.

With the process of people understanding event relations (i.e., learning the content of events), the input of BERT should be event content pairs. This is the first step of input optimization for BERT.

##### 4.3.1. Event restore

In this paper, we call sentences formed by concatenating the extracted three elements of event ET as ETS. If an ETS pair is fed to BERT, the performance of recognition may reduce as losing much semantic information. For instance, the ETSs of  $ET_{7-1}$  (TaiHai Nuclear, paying, dividends) and  $ET_{7-2}$  (the stock market, buy, it) are  $ETS_{7-1}$  (TaiHai Nuclear paying dividends) and  $ETS_{7-2}$  (the stock market buy it), as shown in Fig. 7. It can be seen that neither the adversative conjunction “but” nor the negative word “not” is included in the ETSs. Therefore, in order to improve the recognition effect, Event Restore Sentence (ERS) pair with more semantic information is exploited as the input, which is a further optimization of BERT input. Although the idea is fundamentally straightforward, it is hard to extract structured events and restore them to event sentences with word order for existing methods of event extraction.

Subsequently, we design an approach to generate the event restore sentence for each extracted event and combine them into pairs with originally associated edges. These are the final input of BERT. The corresponding algorithm is shown in Algorithm 1, where nodes are numbered in ascending order of words in an original sentence. Note that the ellipsis nodes are also restored as the omitted event content by the edges established in the completion process in Section 4.2.

Based on the above algorithm, events  $ET_{7-1}$  and  $ET_{7-2}$  are restored to event restore sentences  $ERS_{7-1}$  (TaiHai Nuclear is finally paying dividends) and  $ERS_{7-2}$  (but the stock market does not buy it), respectively (see Fig. 7). Thereinto, the  $ERS_{7-2}$  replenishes the conjunction “but” and negative “not”. The specific differences between ETS and ERS are demonstrated in Fig. 7.

**Algorithm 1** Generation algorithm of Event Restore Sentence pair.

---

**Input:** SSDP of sentence.  
**Output:** ERS list *ersl*, ERS pair list *epl*.

```

1: ersl =  $\emptyset$ ;
2: epl =  $\emptyset$ ;
3: obtain all children eNodes of Root in SSDP;
   //generate event restore sentence for each event
4: for eNode in eNodes do
5:   ers = ""; //event restore sentence
6:   get children cNodes of eNode including ellipsis nodes;
   //order of words in original sentence
7:   sort cNodes by node id in ascending order;
8:   for node in cNodes do
9:     ers += node.tag; //tag is the word attribute of node
10:  add ers to ersl;
11:  if eNode has left brother node lbNode then
12:    seek event restore sentence lb_ERS of lbNode in ersl;
13:    add (lb_ers, ers) into epl;
14: return ersl, epl;

```

---

**4.3.2. Event core embeddings**

Although event restore sentence pairs contain core elements of event ET, they are not explicitly tagged. There is still a barrier for machines to learn semantic relations accurately. Thus, an additional embedding layer is proposed to encode the ET information in BERT. This is an internal embedding optimization of BERT to enhance its encoding ability to inputs.

Original BERT has three embedding layers (i.e., token, segment, and position) and are designed to encode semantics, corresponding sentence, and the position of tokens. The first token of each input sequence is always a special classification token ([CLS]). The final hidden state corresponding to this token is used as the aggregate sequence representation for classification tasks. Each event restore sentence pair is packed together into a sequence, and a special token ([SEP]) is adopted to differentiate event restore sentences in it.

Given a token in the input text, its initial embedding representation is constructed by summing corresponding embeddings of token, segment, and position. Based on this, we augment an event core embedding layer and constitute a four-layer embedding to encode inputs. The visualization of this construction can be seen at the bottom of Fig. 2.

Furthermore, Fig. 2 demonstrates the process from extracted ET in Syntactic Semantic Dependency Parsing graph to Event Core Embeddings with purple lines, where the purple rectangles denote event core elements and light blue ones are non-core. Since BERT is trained as a unit of Chinese characters, the nodes of ET are the results after the word segmentation [32], and a mapping process needs to be designed. The specific operation is as follows. First, obtain event ET. Then, for each element of ET, record its starting and ending positions in the event restore sentence. Last, mark tokens in the starting and ending range as  $Em_c$ , the others as  $Em_{-c}$ , where  $Em_c$  and  $Em_{-c}$  are represented by 1 and 0, respectively.

**4.3.3. Relation classifier**

This paper employs a BERT classifier to identify types of event relations, and the training architecture and loss function are the same as BERT. In Fig. 2, we leave out other modules of BERT and only show the embedding generation process of inputs in BERT. Given the domain characteristic of a corpus, schemes of direct classification and fine-tuning are designed together.

Note that, the Event Core Embeddings operation based on BERT (denoted as BERT[ECE]) is a sub-component of the framework CFERE. BERT[ECE] is used to identify event relation types, and CFERE responds to the whole process, including event extraction, event sentence restoring, and the combination of event sentences. These modules are designed to handle and prepare informative event sentence pairs for event relation type recognition.

**5. Experiments and results**

In the following, we discuss our data, experimental settings, baselines, and key results.

**5.1. Dataset and hyper-parameter**

**Annotated corpus.** To verify the extraction performance of our CFERE framework for event relation types proposed in this paper, we annotated a corpus that includes event pairs and corresponding relations for experiments. The corpus is derived from financial news on Sina Finance (finance.sina.com.cn). We randomly selected 4,312 news headlines containing at least two events.

Three university faculty members with expertise in finance and economics participated as annotators, and they were all Chinese native speakers. The whole process is divided into four phases. (1) The team learned the content and format of annotated corpus and made a few trial annotations (300 headlines). (2) Annotators labeled the public part (1,000 headlines including the trial annotations) and checked the differences between each other. (3) According to the finalized standards, annotators independently marked the allotted portion in the remaining corpus (1,104 headlines for each). (4) Each annotator cross-verified the annotated results of others and discussed different opinions.

**Table 1**  
Statistical results of event relations.

| Event Relations            | Numbers | Event Relations  | Numbers |
|----------------------------|---------|------------------|---------|
| <i>Supplement</i>          | 1,015   | <i>Causality</i> | 854     |
| <i>Degree</i>              | 200     | <i>Parataxis</i> | 1,241   |
| <i>Strategy</i>            | 413     | <i>Contrast</i>  | 361     |
| <i>Phenomenon-Instance</i> | 774     | <i>Other</i>     | 162     |

The standards include: (1) if a sentence is not triggered by a verb, it is considered as non-event; (2) each core verb triggers an event, but multiple events of adjacent core verbs in the same sentence are combined into one event, in which only adverbs between core verbs are also considered to be adjacent; (3) when annotated results are inconsistent, related sentences will be further discussed, and then the annotators vote to decide the final annotation. Note that, the above event identification standards contribute to the construction of event sentence pair and relation recognition, and event information (e.g., subject, predicate, and object) is not annotated.

The annotation is based on the annotator's intuition, which may be very subjective. The reliability of annotations needs to be verified. We assess inter-annotator agreement based on their annotated results of the same 1,000 news headlines. Kappa score ( $k$ ) [33] is widely adopted by computational linguistics to correct for agreement on the reliability of the annotation scheme by chance [34]. The  $k$  statistics is employed to measure inter-annotator agreements for annotations of event relations and is calculated based on Confusion Matrix ( $CM$ ) as follows:

$$k = \frac{P_a - P_e}{1 - P_e}, \quad (1)$$

where  $P_a = \sum_{i=1}^n CM(i, i) / \sum_{i=1}^n \sum_{j=1}^n CM(i, j)$ , that is accuracy, and  $P_e = \sum_{i=1}^n (\sum_{j=1}^n CM(i, j) * \sum_{j=1}^n CM(j, i)) / (\sum_{i=1}^n \sum_{j=1}^n CM(i, j))^2$  (i.e., the sum of the product of actual and predicted numbers for each category divided by the square of the total number of samples, and  $i, j$  are rows and columns of  $CM$ ).

The kappa result in this paper is the average agreement among three annotators, and the final kappa score  $k$  is 92.03%. The reason for the high agreement rate mainly benefits from the trial annotations, in which many different opinions were unified. In total, the corpus contains 9,332 events and 5,021 event pairs. Specific statistics of each type of event relation are reported in Table 1, where the type *Other* ("O" for short) represents cases that do not belong to any of the seven relation types. Experimental code and corpus are available at <https://github.com/hawisdom/CFERE>.

**Experimental settings.** We conducted five independent experiments on the dataset. For each experiment, 30% of the dataset was randomly selected as the test set and executed five times. The average value was taken as the final result. Given the large number of enterprise names and terms in the finance domain, we added all of them to the segmentation dictionary to improve the effectiveness of the DP tool. Besides, since a sentence may contain more than two events and event pairs, to match each event pair to the event relation annotated manually, we split each sentence into multiple single-event sub-sentences, resulting in an event pair per line. The splitting sentences (all lines after splitting) containing event pairs are called *Event Label Sentence* (ELS). Due to partial dependency parsing errors, 294 non-conforming event pairs were filtered during event extraction. For example, practical verbs were not analyzed as verbs, and dependency relations erroneously parsed something other than parallel relation COO. Finally, a total of 4,727 events restore sentence pairs, along with corresponding annotated relations, were poured into BERT for relation identification.

In terms of the model selection, the BERT-Base<sup>3</sup> Chinese pre-trained model was used. All hyper-parameters are consistent with the pre-trained model, in which *epochs*, *batch\_size*, *learning\_rate*, and word embedding dimensions are set to 2, 16, 5e-5, and 768, respectively.

## 5.2. Baselines

Considering that there are few studies similar to ours, and single relation extraction (e.g., causal and temporal relation) focuses on their specific fine-grained issues, they are unsuitable as baselines. Also, we mainly verify the recognition performance of optimized BERT for multi-type event relations. The sophisticated deep learning methods are decent choices that can demonstrate our model better.

We selected CNN, RCNN [35], Bi-LSTM, Bi-LSTM + Attention [36], Adversarial LSTM [37], along with two state-of-the-art language models: GCN (Gated Convolutional Network) [38] and ERNIE [39]. The inputs are event restore sentence pairs generated by the Event Restore module, and word embeddings are learned by model on the corpus derived from 798,262 news posted on Sina.com from 2018 to 2019 [40]. Please note that there is one exception for word embedding; that is, ERNIE has its own embeddings. The final parameter settings of Word2Vec are as follows. Word embedding dimension, *window*, *min\_count*, and training algorithm are 200, 5, 5, and CBOW, respectively. ERNIE adopts the ERNIE 1.0 pre-trained model, and fine-tuning parameters remain unchanged herein. The hyper-parameters of other baselines are set following the suggested values in the original papers.

<sup>3</sup> <https://github.com/google-research/bert>.

**Table 2**  
Performance of seven baselines and CFERE with F1-scores.

| Models              | Sup.         | Deg.         | Str.         | Phe-Ins.     | Cau.         | Par.         | Con.         | O            | Mi-F         | Ma-F1        |
|---------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| CNN                 | 89.22        | 72.34        | 73.18        | 86.83        | 79.49        | 73.44        | 59.00        | 21.07        | 77.66        | 69.32        |
| RCNN                | 87.25        | 55.89        | 57.57        | 83.09        | 76.97        | 69.74        | 32.12        | 9.54         | 72.15        | 59.02        |
| Bi-LSTM             | 88.54        | 68.75        | 74.69        | 88.50        | 81.24        | 74.61        | 63.68        | <b>32.07</b> | 78.64        | 71.51        |
| Bi-LSTM + Attention | 87.77        | 73.28        | 76.40        | 89.10        | 79.98        | 74.03        | 61.16        | 25.75        | 78.19        | 70.94        |
| Adversarial LSTM    | 85.02        | 74.02        | 78.32        | 87.58        | 81.59        | 75.65        | 64.16        | 28.27        | 79.25        | 71.45        |
| GCN                 | 88.55        | 68.53        | 67.93        | 85.68        | 79.09        | 72.32        | 53.64        | 19.35        | 76.04        | 66.88        |
| ERNIE               | <b>92.39</b> | 80.00        | 78.63        | <b>92.27</b> | 80.72        | 77.52        | 70.18        | 8.00         | 81.97        | 72.46        |
| CFERE               | 92.20        | <b>82.12</b> | <b>91.26</b> | 91.94        | <b>81.83</b> | <b>80.99</b> | <b>75.66</b> | 22.66        | <b>84.50</b> | <b>77.33</b> |

Note: Mi-F1 and Ma-F1 are Micro-F1 and Macro-F1, respectively.

**Table 3**  
F1-scores performance of original BERT and optimized BERT on ETS, ERS, and ELS.

| Operations      | Sup.         | Deg.         | Str.         | Phe-Ins.     | Cau.         | Par.         | Con.         | O            | Mi-F1        | Ma-F1        |
|-----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| BERT[ETS]       | 90.87        | 78.06        | 58.57        | 89.85        | 78.79        | 73.44        | 41.57        | 7.19         | 76.65        | 64.79        |
| BERT[ERS]       | 91.75        | 81.30        | 90.93        | 91.66        | 81.16        | 80.62        | 75.51        | 14.85        | 84.06        | 75.97        |
| BERT[ELS]       | 92.16        | 81.51        | 92.71        | 91.80        | 81.93        | 81.37        | 75.70        | 22.01        | 84.71        | 77.40        |
| BERT[ETS + ECE] | 90.87        | 78.18        | 59.97        | 89.98        | 78.82        | 74.01        | 42.65        | 10.27        | 77.06        | 65.59        |
| BERT[ERS + ECE] | 92.20        | 82.12        | 91.26        | 91.94        | 81.83        | 80.99        | 75.66        | <b>22.66</b> | 84.50        | 77.33        |
| BERT[ELS + ECE] | <b>92.28</b> | <b>82.73</b> | <b>92.78</b> | <b>92.15</b> | <b>82.57</b> | <b>81.41</b> | <b>76.63</b> | 22.23        | <b>85.04</b> | <b>77.85</b> |

### 5.3. Key results

The F1-scores of models are reported in Table 2. As can be seen, in terms of both Micro-F1 and Macro-F1, our method is consistently superior over other methods with 2.53~12.35 and 4.87~18.31 percentage points, respectively. Note that our method does not achieve the best performance for the O type. Also, compared with the mainly focused types, the O type has a relatively small sample size, leading to worse performance.

By comparing various baselines, it can be seen that ERNIE, Adversarial LSTM, and Bi-LSTM perform well. This is because the entity mask strategy is added in ERNIE, and the sequence information of input text is encoded by the latter two models. However, although ERNIE considers the entity knowledge, our scheme still outperforms it with a 2.53 and 4.87 percentage points increase in terms of Micro-F1 and Macro-F1. The major reason is the design of Event Core Embeddings (ECE), which explicitly encodes the core content of events. For the GCN (Gated Convolutional Network), given that it focuses on controlling what information is allowed to pass through, and ignores additional structure/knowledge contained in the text, thus the performance is slightly lower than other models. In conclusion, all results consistently suggest that CFERE is the best scheme for financial event relation extraction.

## 6. Discussion

In this section, we provide additional ablation experiments and an experimental extension on other English corpus.

### 6.1. Ablation cases

To confirm the operation of Event Restore Sentence (ERS) and Event Core Embeddings (ECE), we designed experiments on original BERT and optimized BERT (i.e., add ECE), and inputs are the pairs of ETS (i.e., concatenate event ET), ERS, and Event Label Sentence (ELS, that is splitting sentences containing event pairs). The performance comparison is reported in Table 3, where BERT[ERS + ECE] is our scheme. The results confirm that the scheme BERT[ERS + ECE] achieved the best performance (Micro-F1 and Macro-F1 are 84.50% and 77.33%, respectively) under actual scenarios lacking annotations. In the following, we provide further analyses to investigate the performance impacts of (1) original BERT, (2) optimized BERT, and (3) the comparison between the two BERT.

**Original BERT.** Looking at the results in the first three lines in Table 3, we can summarize two important findings. First, the Micro-F1 and Macro-F1 of BERT[ERS] are only 0.65 and 1.43 percentage points lower than the ideal scheme BERT[ELS], indicating that the ideal effect can be approximated by injecting ERS. It also verifies the value of the event restore sentence. The errors of predicate extraction are a possible reason leading to ERS under ELS. Second, the classification effectiveness of BERT[ERS] is significantly higher than BERT[ETS], with Micro-F1 and Macro-F1 increasing by 7.41 and 11.18 percentage points respectively, demonstrating the significance of ERS.

The reasons include: (1) ETS loses learning ability with less contextual information; (2) the extracted error propagation is more prominent for ETS, yet ERS eases. Even though subjects or objects are extracted incorrectly, correct subjects or objects are still in ERS, i.e., the input embedding module of the original BERT does not perceive the errors. Indeed, given the predicate extraction errors, ERS cannot avoid their impact, like ETS.

**Table 4**  
Statistics of event relations in the extended corpus.

| Event Relations    | Original Numbers | Numbers Preserving One Relation |
|--------------------|------------------|---------------------------------|
| <i>Inheritance</i> | 67               | 56                              |
| <i>Expansion</i>   | 120              | 106                             |
| <i>Contingency</i> | 339              | 266                             |
| <i>Comparison</i>  | 15               | 11                              |
| <i>Temporality</i> | 796              | 548                             |

**Table 5**  
F1-scores performance of various event relations in English corpus.

| Model | <i>Inher.</i> | <i>Expan.</i> | <i>Cont.</i> | <i>Temp.</i> | <i>Disrelated</i> | Micro-F1 | Macro-F1 |
|-------|---------------|---------------|--------------|--------------|-------------------|----------|----------|
| CFERE | 95.74         | 85.11         | 64.63        | 75.94        | 80.92             | 80.89    | 80.47    |

**Optimized BERT.** Comparing the results in the bottom three lines in Table 3, the following three findings are noticeable. First, the performance of BERT[ERS + ECE] is still very close to the ideal scheme BERT[ELS + ECE], with only 0.54 and 0.52 percentage points less for Micro-F1 and Macro-F1, respectively. The consistent results show that ERS not only retains more semantic information, but also filters the impact caused by the extraction errors of event core components, indicating the robustness of ERS.

Second, BERT[ERS + ECE] significantly improves Micro-F1 and Macro-F1 by 7.44 and 11.74 percentage points, respectively. Last, regarding the performance of each event relation in BERT[ERS + ECE], *Supplement*, *Degree*, *Strategy*, *Phenomenon-Instance* performed well, whose F1 values are all over 90%. A possible reason is that they contain more domain information, such as the extensive use of numerical words and trend verbs, the description of enterprises, and their subordinate fields, etc. Due to the universality in different domains and the challenges in the semantic comprehension of sentences, the event relation extractions for *Causality*, *Parataxis*, and *Contrast* are relatively more complex tasks. However, our method still resulted in F1 over or about 80%, confirming the excellent overall performance of the CFERE in identifying multi-type event relations.

**Optimized BERT vs. Original BERT.** For ETS, ERS, and ELS, the performance of optimized BERT outperforms the original BERT, whose Micro-F1 and Macro-F1 raise (ETS) 0.41 and 0.80, (ERS) 0.44 and 1.36, and (ELS) 0.33 and 0.45 percentage points, respectively. The average increases are 0.39 and 0.87 percentage points, which confirms the effectiveness of ECE in improving the ability of feature learning.

Moreover, compared with the increase in BERT[ERS + ECE] over BERT[ERS], the increase in BERT[ETS + ECE] is less than that in BERT[ETS], since ECE has some side effects while playing a positive role in ET extraction errors. The impact of BERT[ETS + ECE] is reflected in ETS and ECE, respectively, while BERT[ERS + ECE] is mainly reflected in ECE. That is, ERS is less affected by extraction errors of core components of events, mitigating the negative impact of ECE to a certain extent.

In summary, the results confirm that ERS is the best input and ECE facilitates event relation identification. Also, the events extracted by the Syntactic Semantic Dependency Parsing graph are reliable for event relation recognition.

## 6.2. Experimental extension

Although the model we developed is based on the characteristics of Chinese texts, considering that the model is capable of handling complex sentence structures, our model should be utilized to deal with English texts. Due to verifying language flexibility, we only performed the experimental extension based on our CFERE framework.

**Additional dataset.** To evaluate the performance of our method when being utilized with English corpus, the corpus annotated by literature [13] is tested, which aims to build cross-document event relations and bias hierarchical relations between events. The data source is news documents from ACE2005<sup>4</sup> corpus and some news documents about the “Malaysia Airlines 17” incident. It includes five main types of event relations: *Inheritance*, *Expansion*, *Contingency*, *Comparison*, and *Temporality*, along with *Disrelated* and *None* relation types. Most of them are *None* with 19,175 event pairs. Also, in the dataset, all relations of each event pair are annotated. Since *Temporality* is usually at the top of each relation list (each sentence corresponds to an event relation list), we attained the last event relation in each relation list to balance the number of event relations. There are 987 event pairs with five types and 661 *Disrelated* types. Important data statistics are summarized in Table 4.

**Experimental Details and Results.** In this experiment, the *Comparison* was eliminated because of the small number. Also, to avoid data imbalance, we randomly selected 300 samples for each relation type. In particular, the over-sampling method is adopted for *Inheritance*, *Expansion*, and *Contingency*, under-sampling is processed for *Temporality* and *Disrelated*. We used the Stanford CoreNLP tool for dependency syntactic and BERT-Base (Uncased) pre-trained model as the classifier. F1-scores of these relations are reported in Table 5. The results demonstrate the fine performance of our method in handling multi-type relation recognition tasks on the English corpus, confirming our method’s flexibility when applied to other languages.

<sup>4</sup> <https://catalog.ldc.upenn.edu/LDC2006T06>.

## 7. Conclusion

This paper addressed the challenging multi-type event relation extraction problem. Based on the existing causality type, we first defined six types of event relations in the financial field. Then, the Core Verb Chain and Syntactic Semantic Dependency Parsing graph were employed to identify and extract events in sentences. Meanwhile, the associations between events and ellipsis completion of event elements were implemented. Also, we strengthened the original BERT to enhance its performance by inputting Event Restore Sentence pairs and adding internal Event Core Embeddings. Last, we fine-tuned it on our financial corpus for final type recognition of event relations. The experimental results verify the effectiveness and flexibility of our framework.

## CRedit authorship contribution statement

**Qizhi Wan:** Conceptualization, Methodology, Software, Writing – original draft. **Changxuan Wan:** Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing. **Keli Xiao:** Validation, Writing – original draft. **Rong Hu:** Data curation, Software. **Dexi Liu:** Funding acquisition, Resources, Writing – review & editing. **Xiping Liu:** Funding acquisition, Resources, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

I have share the link to my data and code in the revised manuscript.

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